MIL-I-43639A
20 May 1971
SUPERSEDING
MIL-I-43639
2 May 1969

MILITARY SPECIFICATION

INK, MARKING, NAMETAPE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. Scope

1.1 This specification covers one type of ink for application on nylon tapes (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-S-571	-	Sodium Carbonate, Anhydrous, Technical.
O-S-604	-	Sodium Metasilicate, Technical.
P-S-579	-	Soap, Laundry, Chip and Powdered.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.
PPP-B-676	-	Boxes, Setup.

MILITARY

MIL-T-43664	-	Tape, Textile, Nylon (For Name Tapes).

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

FSC 7510

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:54Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48", Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF HEALTH EDUCATION AND WELFARE

FOOD AND DRUG ADMINISTRATION

Federal Hazardous Labeling Act

(Application for copies should be addressed to the Division of Public Information, Food and Drug Administration, Washington, D.C. 20203.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INCORPORATED, AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:54Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Standard sample.- The ink, when applied to one inch of OD No. 7 nylon tape (see 4.4.1), shall match the standard sample for color and legibility and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.2 Materials.- The manufacturer is given a wide latitude in the selection of materials and manufacturing processes, provided the ink meets all the requirements of this document. The ink shall be a heavy body liquid or paste, ready for use from the container, for letter press printing.

3.3 Prohibited materials.- The ink shall contain no benzene, aniline, or chlorinated solvents.

3.4 Color.- Unless otherwise specified, (see 6.2), the color of the ink shall be black, and when applied to the specified tape, the impressions shall be dull black equal to color chip No. 37038 of FED-STD-595, when tested as specified in 4.3.4.

3.5 Legibility.- The marking ink shall produce a sharp, legible impression on nylon tape when tested as specified in 4.3.4.

3.6 Drying.- The ink shall set to touch in 5 minutes and shall not smear, when rubbed lightly, 15 minutes after application when tested as specified in 4.3.4.

3.7 Fastness and legibility of the ink to laundering.- The ink, when applied on the specified nylon tape (see 4.4.1), shall have equal or better fastness and legibility than the standard sample after being washed three times when tested as specified in 4.3.4.

3.8 Labeling.- Labeling shall conform to the regulations under the Federal Hazardous Substances Act, Part 191. In addition, each tube and interior box (see 5.1.1) shall be clearly and legibly lithographed, stenciled or printed with a non-corrosive weather resistant ink which will not smear nor rub off during handling. The label data shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 20260-21T21:54Z
Check the source to verify that this is the current version before use.

DANGER! HAZARDOUS LIQUID RAPIDLY ABSORBED THROUGH SKIN CAUSES SEVERE BURNS (These 2 lines shall be in minimum 10-point type)

Do not get in eyes, on skin, on clothing.

Avoid breathing vapor.

Do not take internally.

In case of contact, immediately remove all contaminated clothing, including shoes, and flush skin or eyes with plenty of water for at least 15 minutes; for eyes, get medical attention.

Wash clothing before re-use.

This ink is to be used for marking nylon name tapes.

Date of packaging.

Thin this ink with _____.

(Manufacturer shall furnish name of product suitable for thinning this ink.)

Antidotes: If accidentally swallowed, administer milk or raw eggs and give large amounts of water. Then cause vomiting by inserting finger down throat.

3.9 Workmanship.- The finished product shall be free from foreign matter.

3.10 Storage life.- The ink, after storage in the original tube for a period of not less than one year, shall be free from grit, lumps, and skins, and shall not gel, settle, harden or otherwise deteriorate, when tested as specified in 4.3.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any

Source: https://assist.dla.mil -- Downloaded: 202640-21T21:54Z
Check the source to verify that this is the current version before use.

other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Inspection of materials and components.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3 Inspection of the end item.-

4.3.1 Examination of the end item.- The end item shall be examined for the defects in accordance with the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth in 4.3.3. The lot shall be expressed in units of filled unit containers of the specified capacity for the examination in 4.3.1.1 and 4.3.1.2;in units of shipping containers for the examination in 4.3.2.1, and in units of pallets for the examination in 4.3.2.2.

4.3.1.1 Examination of unit container.- The sample unit for this examination shall be one filled container.

Examine	Defects
Container	Not size, shape, type container specified. Any evidence of seepage or leakage from any part of the container including closure.
Contents	Not color specified. Presence of foreign matter.

4.3.1.2 Examination of filled container for net contents.- The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container for all sample units examined is less than specified or indicated.

4.3.2 Examination of preparation for delivery requirements.-

4.3.2.1 Examination for packaging, packing and marking.- An examination shall be made to determine compliance with the packaging, packing and marking

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:54Z
Check the source to verify that this is the current version before use.

requirements of Section 5. Defects shall be as listed below. The sample unit for this examination shall be one shipping container fully prepared for delivery except that it shall not be palletized and need not be sealed. Shipping containers fully prepared for delivery that have not been palletized shall be examined for defects of closure. The lot shall be the number of shipping containers in the end item inspection lot.

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of packages per shipping container is less than specified.

4.3.2.2 Examination for palletization.- An examination shall be made to determine compliance with the palletization requirements of Section 5. Defects shall be as listed below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot.

Examine	Defects
Finished dimension	Length, width or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Weight	Exceeds maximum load limits.

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:54Z
Check the source to verify that this is the current version before use.

4.3.3 Inspection levels and acceptable quality levels (AQL's) for examinations.- The inspection levels, for determining the sample size, and the acceptable quality levels (AQL's), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	I	2.5
4.3.1.2	S-2	N.A.
4.3.2.1	S-2	2.5
4.3.2.2	S-1	6.5

4.3.4 Testing of the end item.- The end item shall be tested for the characteristics specified in table I. The lot shall be expressed in terms of 8 ounce tubes. The sample unit for testing shall be a one pint composite. The composite shall be made by thoroughly mixing equal portions selected at random throughout the lot in the frequency listed below. The composite shall be placed in a clean, dry glass container. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if the composite fails to meet any test requirement specified.

Lot size (tubes)	Number of portions to form composite
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:54Z
Check the source to verify that this is the currentversion before use.

INSTRUCTIONS FOR TESTING OF THE END ITEM

TABLE I

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver	Per composite unit	Pass or Fail	Numerically to Nearest		
Prohibited material	3.3	1/							
Color	3.4	4.4.1			2	X			
Legibility	3.5	4.4.1			2	X			
Drying	3.6	4.4.2			2	X			
Fastness, and legibility of the ink after laundering	3.7	4.4.3			2	X			
Storage life	3.10	1/							

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the requirement stated.

AMXRE Form 598
8 February 1965

MIL-I-43639A

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:54Z
Check the source to verify that this is the current version before use.

4.4 Test procedures.-

4.4.1 Color and legibility.- The ink shall be applied to a one inch OD No. 7 nylon tape conforming to MIL-T-43664. The printing type shall be Franklin Gothic Condensed 48 point, with 1/2 inch letters. Print in one stroke the first ten letters of the alphabet on the tape, using a printing press and applying sufficient ink to obtain a uniform solid opaque impression. After one hour, observe the color and the sharpness of the letters. The dry print shall be compared with color chip No. 37038 for color and dullness.

4.4.2 Drying.- Apply the test ink as specified above. After five minutes, touch the printed letters with a finger and examine for transfer of ink. The ink is set to touch if there is no transfer. After fifteen minutes rub letters with a finger, and observe smearing.

4.4.3 Fastness and legibility of the ink after laundering.- 3.5 $\pm$ 0.3 grams of the printed nylon tape (see 4.4.1) shall be laundered 3 times in accordance with the following procedure.

4.4.3.1 Apparatus.- Accelerator or similar machine (see 6.3) capable of varying the speed of the rotor from 0 to 4000 revolutions per minute with a 4-1/2 inch "S" rotor. The liner from the test chamber shall be rubber ribbed, having 2 ribs per inch, with the ribs approximately 1/16 inch in height.

Soaking chamber.- Any convenient vessel for soaking the test samples.

Steam pressing.- A flat-bed press maintained at a temperature of 275° to 300° F., having hot head with polished metal top.

Hand iron.- A hand iron weighing approximately 6 pounds, and maintained at a temperature of 275° to 300° F.

4.4.3.2 Reagents.-

Water.- The water used in the preparation of the soap solution shall have a water hardness of not more than 50 parts per million, expressed as calcium carbonate.

Soap - alkali solution.- The soap - alkali solution shall contain 0.2 percent laundry soap, conforming to type I of P-S-579, 0.2 percent sodium metasilicate, conforming to type I of O-S-604 and 0.7 percent sodium carbonate conforming to type I of O-S-571.

Source: https://assist.dla.mil -- Downloaded: 202640-21T21:54Z
Check the source to verify that this is the currentversion before use.

4.4.3.3 Procedure.- When a standard sample has been established, one specimen from the standard sample shall be tested at the same time and under the same conditions as the specimen from the sample unit of material being tested. The test sample and the standard sample shall be immersed in the soap - alkali solution for 30 ± 2 minutes at 150° F. at a ratio of 100 milliliters of solution per gram of material. The accelerator or similar machine shall have a speed control setting of approximately 50, and the motor adjusted to 200 revolutions per minute. The test samples are placed in the accelerator with 150 milliliters of the soap - alkali solution and agitated for 6 minutes ± 10 seconds and at a temperature of 80° ± 2° F. At the end of the laundering cycle, the test samples are removed from the accelerator and rinsed thoroughly in running water at a temperature of 105° ± 5° F. for a period of 5 minutes. The test samples shall then be extracted or passed between wringer rolls to remove excess water and then pressed between clean white cotton cloths using a flat-bed press or hand iron at a temperature of 275° to 300° F. After the third cycle, the test sample shall be examined for fastness and legibility by viewing the samples under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin.

4.4.3.4 Evaluation.-

4.4.3.4.1 Standard sample.- The test specimen will be compared to the specimen of the standard sample and rated as follows: Specimen. Pass: Equal or superior to the standard sample. Fail: Inferior to the standard sample.

4.4.3.5 Report.-

4.4.3.5.1 Fastness of the ink to laundering.- Fastness of the ink to laundering shall be reported as "pass" or "fail" when compared with the standard sample. If the test specimen shows failure when compared to the standard sample, the specimen shall be reported as failing. When failure is reported, the nature of the failure shall also be reported.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Ink shall be furnished in 8-ounce tubes which shall be securely closed with a non-corrosive metal or plastic screw caps. Each

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:54Z
Check the source to verify that this is the current version before use.

tube shall then be packaged in a snug-fitting folding or setup paperboard box conforming to variety 1, style I, type A or B, class a or b, or style II, type D or E, class a or e of PPP-B-566, or type I or II, variety 1, class A, style 2 of PPP-B-676. Each box shall be closed in accordance with the appendix of the applicable box specification.

5.1.2 Level C.- Ink shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.- One hundred 8-ounce tubes of ink, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to grade V2s, style RSC of PPP-B-636. Closure, waterproofing and reinforcing shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B.- One hundred 8-ounce tubes of ink, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type CF, variety SW, class domestic, grade 200, style RSC of PPP-B-636. Closure shall be in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification.

5.2.3 Level C.- Ink, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Palletization.- Unless otherwise specified (see 6.2), ink packed as specified in 5.2, shall be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.- In addition to any special marking required by the contract or order, tubes, interior boxes, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:54Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use.- The ink covered in this specification is a heavy body liquid or paste product to be used for obtaining durable markings on nylon tapes (name tapes) using a letter press printer.

6.2 Ordering data.- Procurement documents shall specify the following:

(a) Title, number and date of this specification.
(b) Color, when applicable (see 3.4).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(e) When palletization is not required (see 5.3).

6.3 Apparatus.- A machine of the type described in this method (see 4.4.3.1) may be obtained from Atlas Electric Devices Co., 4114 N. Ravenswood Ave., Chicago, Ill. 60613.

6.4 Standard samples.- The standard samples may be obtained from the Commanding General, U.S. Army Natick Laboratories, Natick, Mass. Attn: Chief, Chemicals, Leather and Paper Engineering Branch, Clothing and Personal Life Support Equipment Laboratory.

Custodians:

Army - GL
Air Force - 84

User activities:

Army - ME
Navy - MC, SA, PE

Preparing activity:

Army - GL

Project No. 7510-0238

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:54Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Ink, Marking, Nametape MIL-I-43639A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 64 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:54Z
Check the source to verify that this is the current version before use.